

Лабораторная работа №8

Настройка сетевых сервисов DHCP и DNS

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Цель работы

Приобретение практических навыков настройки
DHCP и DNS в локальной сети Cisco Packet Tracer

Настройка DNS-сервера

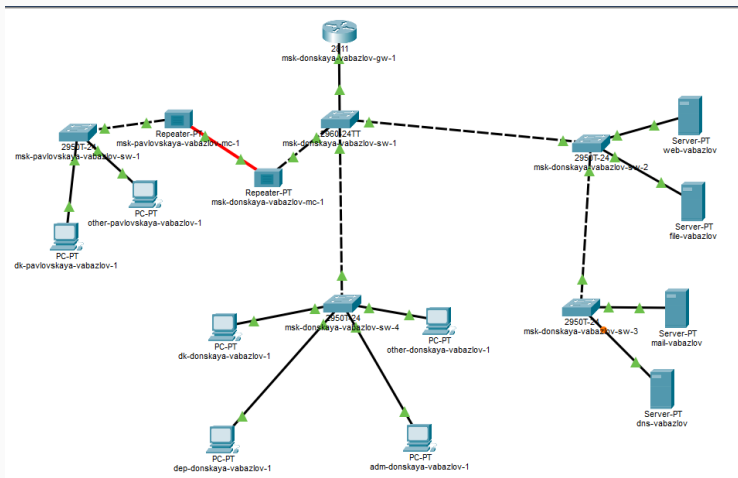


Рис. 1: Схема сети

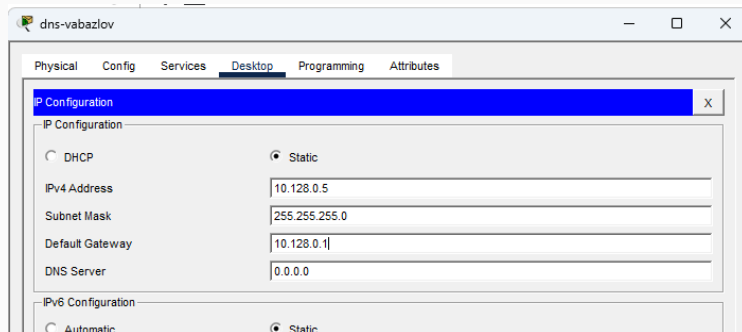


Рис. 2: IP DNS

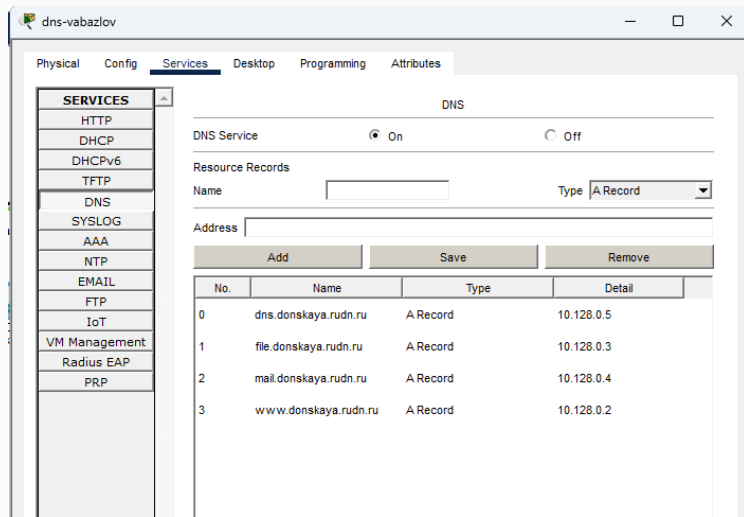
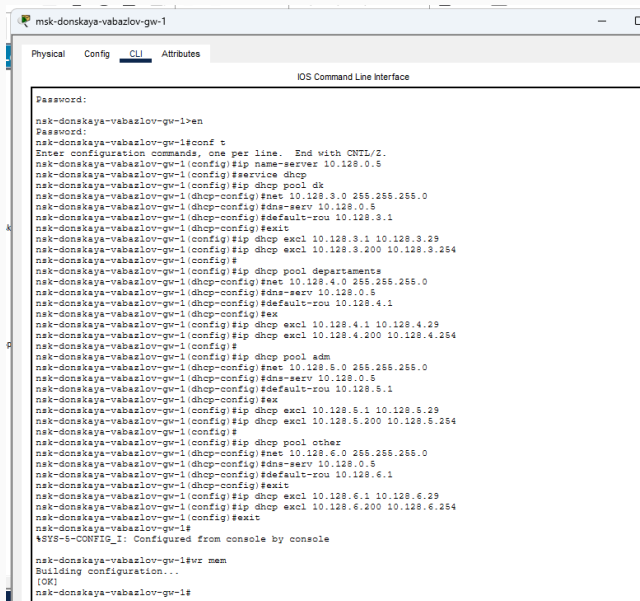


Рис. 3: DNS Records

Настройка DHCP

Конфигурация маршрутизатора



```
msk-donskaya-vabazlov-gw-1
Physical Config CLI Attributes
IOS Command Line Interface

Password:
msk-donskaya-vabazlov-gw-1>en
Password:
msk-donskaya-vabazlov-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-vabazlov-gw-1(config)#ip name-server 10.128.0.5
msk-donskaya-vabazlov-gw-1(config)#service dhcp
msk-donskaya-vabazlov-gw-1(config)#ip dhcp pool dk
msk-donskaya-vabazlov-gw-1(dhcp-config)#net 10.128.3.0 255.255.255.0
msk-donskaya-vabazlov-gw-1(dhcp-config)#dns-serv 10.128.0.5
msk-donskaya-vabazlov-gw-1(dhcp-config)#default-rou 10.128.3.1
msk-donskaya-vabazlov-gw-1(dhcp-config)#exit
msk-donskaya-vabazlov-gw-1(config)#ip dhcp excl 10.128.3.1 10.128.3.29
msk-donskaya-vabazlov-gw-1(config)#ip dhcp excl 10.128.3.200 10.128.3.254
msk-donskaya-vabazlov-gw-1(config)#
msk-donskaya-vabazlov-gw-1(config)#ip dhcp pool departaments
msk-donskaya-vabazlov-gw-1(dhcp-config)#net 10.128.4.0 255.255.255.0
msk-donskaya-vabazlov-gw-1(dhcp-config)#dns-serv 10.128.0.5
msk-donskaya-vabazlov-gw-1(dhcp-config)#default-rou 10.128.4.1
msk-donskaya-vabazlov-gw-1(dhcp-config)#ex
msk-donskaya-vabazlov-gw-1(config)#ip dhcp excl 10.128.4.1 10.128.4.29
msk-donskaya-vabazlov-gw-1(config)#ip dhcp excl 10.128.4.200 10.128.4.254
msk-donskaya-vabazlov-gw-1(config)#
msk-donskaya-vabazlov-gw-1(config)#ip dhcp pool adm
msk-donskaya-vabazlov-gw-1(dhcp-config)#net 10.128.5.0 255.255.255.0
msk-donskaya-vabazlov-gw-1(dhcp-config)#dns-serv 10.128.0.5
msk-donskaya-vabazlov-gw-1(dhcp-config)#default-rou 10.128.5.1
msk-donskaya-vabazlov-gw-1(dhcp-config)#ex
msk-donskaya-vabazlov-gw-1(config)#ip dhcp excl 10.128.5.1 10.128.5.29
msk-donskaya-vabazlov-gw-1(config)#ip dhcp excl 10.128.5.200 10.128.5.254
msk-donskaya-vabazlov-gw-1(config)#
msk-donskaya-vabazlov-gw-1(config)#ip dhcp pool other
msk-donskaya-vabazlov-gw-1(dhcp-config)#net 10.128.6.0 255.255.255.0
msk-donskaya-vabazlov-gw-1(dhcp-config)#dns-serv 10.128.0.5
msk-donskaya-vabazlov-gw-1(dhcp-config)#default-rou 10.128.6.1
msk-donskaya-vabazlov-gw-1(dhcp-config)#exit
msk-donskaya-vabazlov-gw-1(config)#ip dhcp excl 10.128.6.1 10.128.6.29
msk-donskaya-vabazlov-gw-1(config)#ip dhcp excl 10.128.6.200 10.128.6.254
msk-donskaya-vabazlov-gw-1(config)#exit
msk-donskaya-vabazlov-gw-1#
*SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-vabazlov-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-vabazlov-gw-1#
```

- dk — 10.128.3.0/24
- departments — 10.128.4.0/24
- adm — 10.128.5.0/24
- other — 10.128.6.0/24

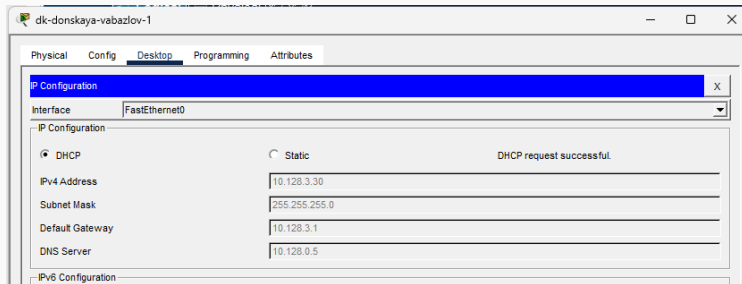


Рис. 5: DHCP pool

Таблица привязок

```
nsk-donskaya-vabazlov-gw-1#
nsk-donskaya-vabazlov-gw-1#sh ip dhcp pool

Pool dk :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)         : 0 / 0
Total addresses                   : 254
Leased addresses                  : 2
Excluded addresses                : 8
Pending event                     : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.128.3.1        10.128.3.1      - 10.128.3.254      2   / 8   / 254

Pool departaments :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)         : 0 / 0
Total addresses                   : 254
Leased addresses                  : 1
Excluded addresses                : 8
Pending event                     : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.128.4.1        10.128.4.1      - 10.128.4.254      1   / 8   / 254

Pool adm :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)         : 0 / 0
Total addresses                   : 254
Leased addresses                  : 1
Excluded addresses                : 8
Pending event                     : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.128.5.1        10.128.5.1      - 10.128.5.254      1   / 8   / 254

Pool other :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)         : 0 / 0
Total addresses                   : 254
Leased addresses                  : 2
Excluded addresses                : 8
Pending event                     : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.128.6.1        10.128.6.1      - 10.128.6.254      2   / 8   / 254
nsk-donskaya-vabazlov-gw-1#
```

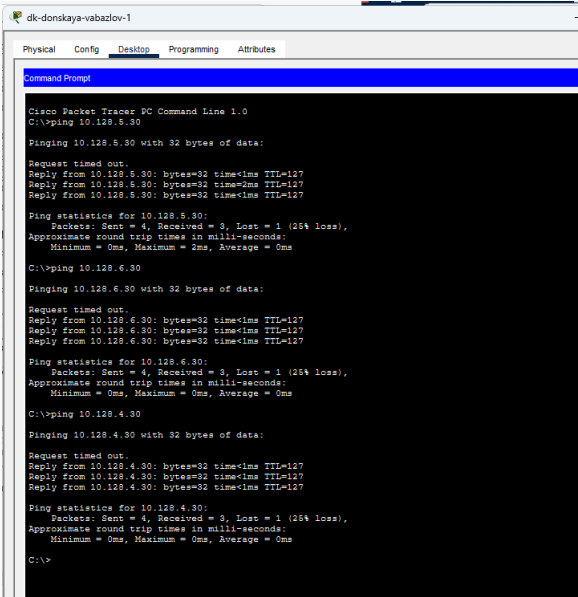
Получение адресов

```
nsk-donskaya-vabazlov-gw-1#  
nsk-donskaya-vabazlov-gw-1#sh ip dhcp bin
```

IP address	Client-ID/ Hardware address	Lease expiration	Type
10.128.3.30	0090.21E7.A67C	--	Automatic
10.128.3.31	0001.9674.D7CE	--	Automatic
10.128.4.30	00E0.B09A.6776	--	Automatic
10.128.5.30	0001.C97D.3A0B	--	Automatic
10.128.6.30	0001.6472.D003	--	Automatic
10.128.6.31	0001.42BA.6960	--	Automatic

```
nsk-donskaya-vabazlov-gw-1#
```

Рис. 7: DHCP client



```
dk-donskaya-vabazlov-1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.5.30

Pinging 10.128.5.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.5.30: bytes=32 time<1ms TTL=127
Reply from 10.128.5.30: bytes=32 time=2ms TTL=127
Reply from 10.128.5.30: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.5.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 10.128.6.30

Pinging 10.128.6.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.6.30: bytes=32 time<1ms TTL=127
Reply from 10.128.6.30: bytes=32 time<1ms TTL=127
Reply from 10.128.6.30: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.6.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.4.30

Pinging 10.128.4.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.4.30: bytes=32 time<1ms TTL=127
Reply from 10.128.4.30: bytes=32 time<1ms TTL=127
Reply from 10.128.4.30: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.4.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Работа DHCP (Simulation)

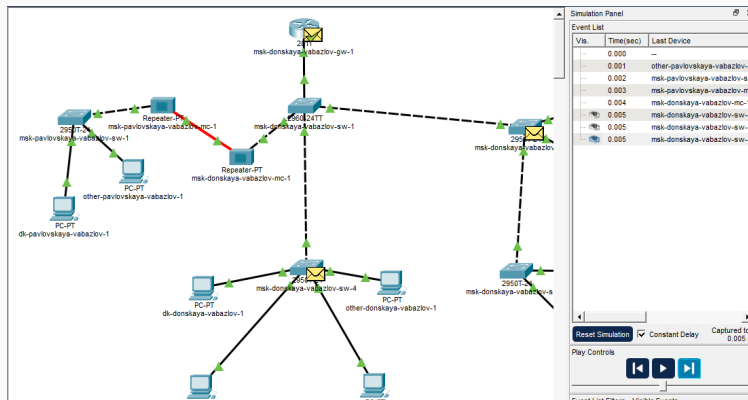


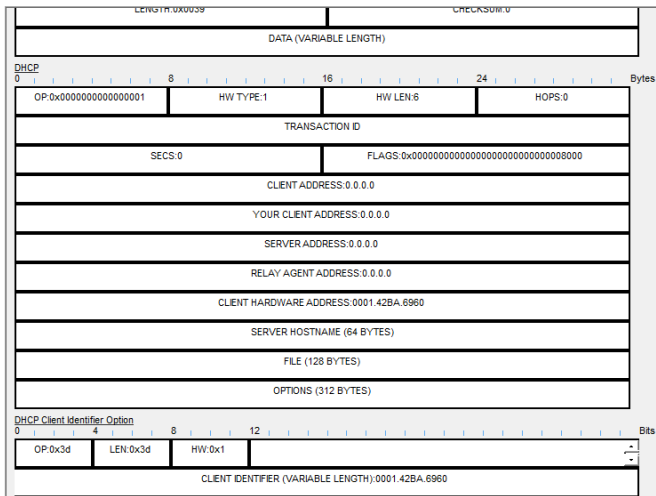
Рис. 9: Simulation

DHCP Discover

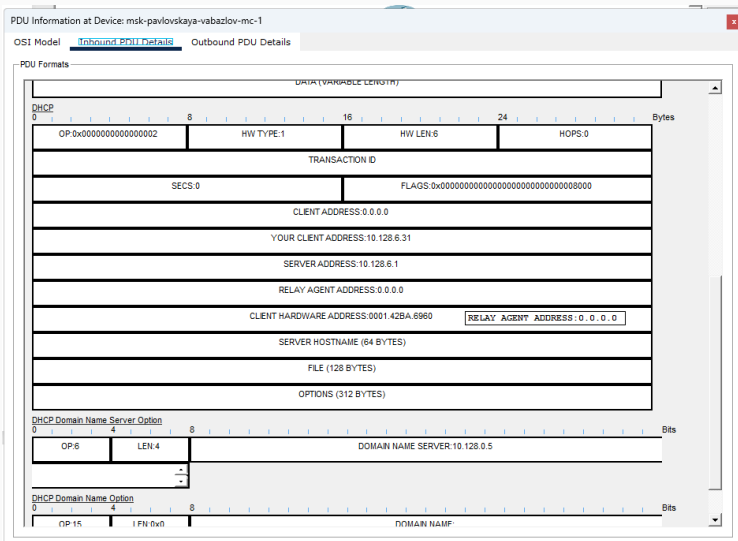
PDU Information at Device: msk-pavlovskaya-vabazlov-sw-1

OSI Model [Inbound PDU Details](#) Outbound PDU Details

PDU Formats



DHCP Offer



DHCP Request

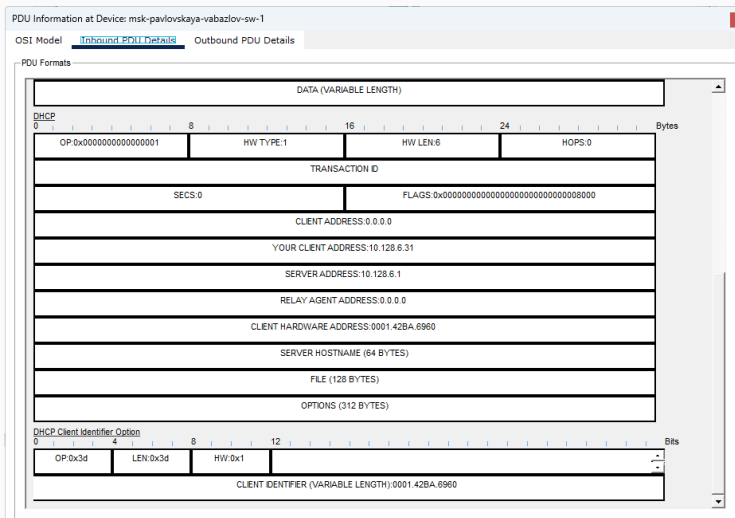
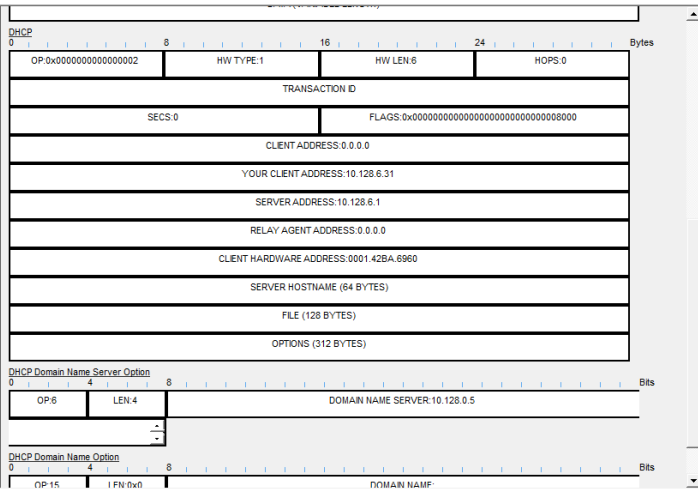


Рис. 12: Request

PDU Information at Device: msk-pavlovskaya-vabazlov-mc-1

OSI Model [Inbound PDU Details](#) Outbound PDU Details

PDU Formats



Итоги работы

- настроен DNS-сервер
- реализовано разрешение имён
- настроен DHCP для нескольких подсетей
- проверена автоматическая выдача адресов
- исследован процесс DHCP в Simulation

Получены навыки настройки сетевых сервисов
и анализа работы протоколов DHCP и DNS